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Foldable Impact-Absorbing Bicycle Helmet

ABSTRACT

A bicycle helmet includes a plurality of rigid shell segments connected by flexible hinge members, allowing the helmet to collapse from an expanded, wearable configuration into a flattened storage configuration. An internal locking mechanism secures the segments in the expanded configuration during use and releases upon actuation of a manual latch to permit folding.

BACKGROUND

Conventional bicycle helmets maintain a fixed rigid shape at all times, occupying significant volume when not worn and making them inconvenient to carry or store in bags, lockers, or bicycle-sharing docking stations. A collapsible helmet design that preserves impact protection while reducing stored volume addresses this inconvenience.

SUMMARY

In one embodiment, the helmet shell is divided into a crown segment and a plurality of side segments joined by living-hinge connectors molded from an elastomeric polymer. A spring-loaded latch, accessible at the rear of the helmet, locks the segments into an expanded dome shape that meets standard impact-absorption requirements. Releasing the latch allows the side segments to rotate inward about the hinges, reducing the helmet's stored thickness by more than half.

CLAIMS

1. A foldable helmet comprising: a crown segment; a plurality of side segments each coupled to the crown segment by a flexible hinge; and a latch mechanism configured to releasably secure the side segments in an expanded configuration. 2. The helmet of claim 1, wherein the flexible hinge comprises an elastomeric living hinge integrally molded with the crown segment. 3. The helmet of claim 1, wherein the latch mechanism is spring-loaded and manually releasable.